CHAPTER 7

Demand Forecasting in a Supply Chain

Instructor

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Session

Week **6**

Chapter Overview & Learning Objectives

Chapter Overview

Supply Chain Management: Strategy, Planning, and Operation (Sunil Chopra, Pearson, 7th edition)

Lecture Material for Supply Chain Management, by H. Y. Jeong

7.1. The Role of Forecasting in a Supply Chain

The basis for all strategic and planning decisions in a supply chain

Examples

Competitive strategy

Network design: facility location and capacity, demand allocation

Production: Purchasing, inventory, production planning, scheduling

Distribution: Transportation planning, warehousing

- The accurate demand forecast enables supply chains to be both **more responsive and efficient**
- Forecasts are always wrong, so should include **expected value and measure of error**
- Long-term forecasts are usually less accurate than short-term forecasts

7.2. Components of a Forecast and Forecasting Methods

Three types of forecasting methods for supply chain planning

Qualitative Methods
Subjective & Judgment-based

Primarily subjective and relies on judgment and opinion. Used when historical data is limited or for new products.

- Expert opinion surveys
- Delphi method
- Market research

Time Series Methods
Historical data analysis

Use historical demand data only. Assumes past patterns will continue into the future.

- Moving average
- Exponential smoothing
- Trend analysis

Causal Methods
Relationship-based

Use the relationship between demand and factors like price, temperature, etc.

- Regression analysis
- Econometric models
- Leading indicators

Combined Forecasting: When companies use quantitative forecasting methods, they must include human input (i.e., subjective factors) when they make their final forecast.

7.2. Components of a Forecast and Forecasting Methods

Components of observed demand and their relationships

Observed Demand Formula

Observed demand (O) = **Systematic component (S)** + **Random component (R)**

The systematic component represents expected value of demand, while the random component deviates from the systematic component.

Level (L)

Current (or t = 0) deseasonalized demand

`L = Deseasonalized demand at t=0`

Trend (T)

Rate of growth or decline in demand

`T = (Dt - Dt-1) / (t - (t-1))`

Seasonality (S)

Predictable seasonal fluctuation in demand

`S = Seasonal factor for period t`

Forecast Error: Difference between forecast and actual demand. **Systematic component = Expected value of demand**, **Random component = Part of forecast that deviates from systematic component**.

7.2. Components of a Forecast and Forecasting Methods

Four key points to be incorporated into the forecasting process

1 Forecast with Cooperative Effort

Forecast with cooperative effort throughout the supply chain and share the forecasts to all relevant planning activities. Cross-functional forecasting, collaborative forecasting.

2 Identify Major Factors

Identify major factors on the demand, supply, and product sides that influence the demand forecast. e.g.) Seasonality, lead time, complementation, substitutability.

3 Forecast at Appropriate Level

Forecast at the appropriate level of aggregation. Consider the granularity needed for decision-making.

4 Establish Performance Measures

Establish performance and error measures for the forecast. Monitor forecast accuracy regularly.

Key Insight: Effective forecasting requires collaboration across the supply chain, consideration of multiple factors, appropriate aggregation levels, and continuous performance monitoring.

7.3. Time-Series Forecasting Methods

General equations for calculating the systematic component

Multiplicative Model

$F = L \times T \times S$

Where **F** = Forecast, **L** = Level, **T** = Trend, **S** = Seasonal factor

Additive Model

$F = L + T + S$

Where **F** = Forecast, **L** = Level, **T** = Trend, **S** = Seasonal factor

Mixed Model

$F = (L + T) \times S$

Where **F** = Forecast, **L** = Level, **T** = Trend, **S** = Seasonal factor

Variable Definitions

- **L (Level)**
  Estimate of level for Period 0 - the base demand without trend or seasonality
- **T (Trend)**
  Estimate of trend - the rate of growth or decline in demand over time
- **S (Seasonal Factor)**
  Seasonal factor - predictable seasonal fluctuation in demand

7.3. Time-Series Forecasting Methods

We will forecast the demand of Tahoe Salt Co. for period 13 - 16

Example: Tahoe Salt Co. (Quarterly Demand)

We will forecast the demand of Tahoe Salt Co. for period 13 - 16

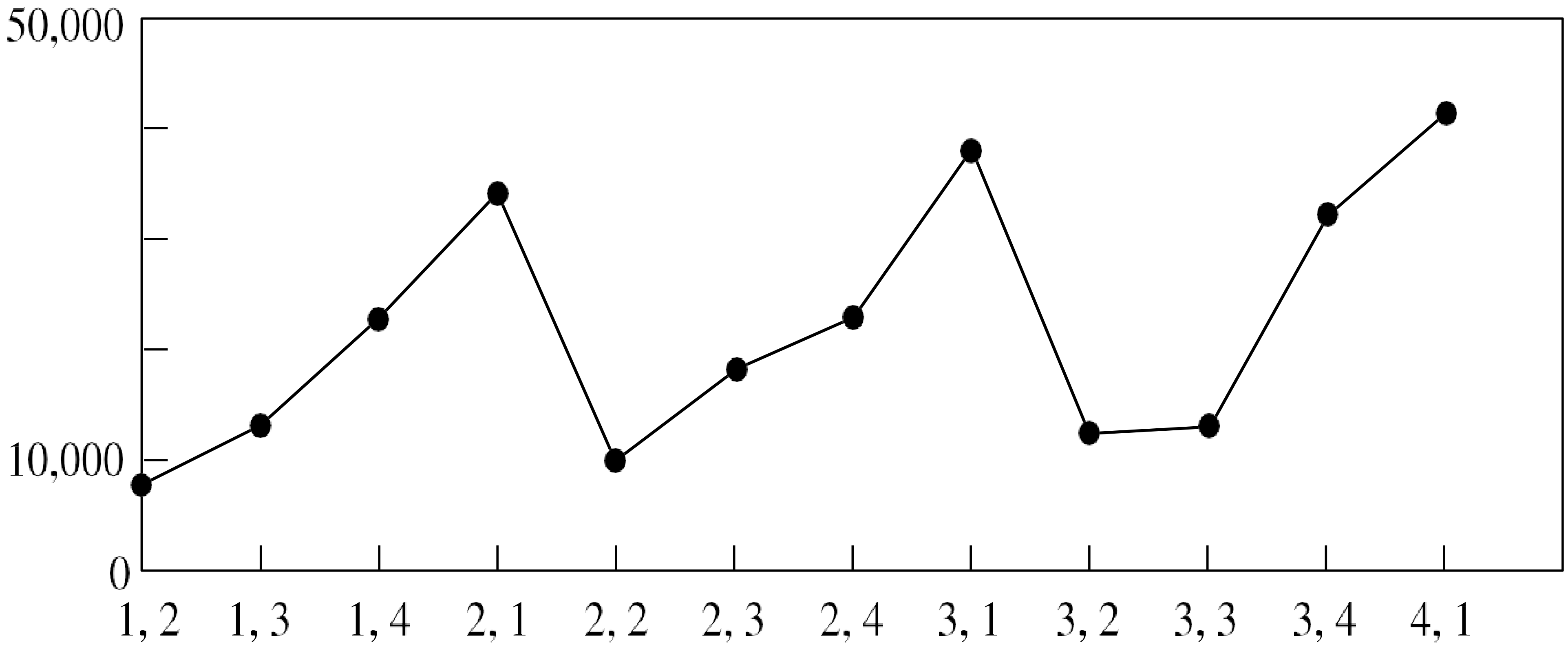
Year	Quarter	Period t	Demand Dt	Season ID	Season mapping note
1	Q2	1	8,000	1	Q2 -> Season 1
1	Q3	2	13,000	2	Q3 -> Season 2
1	Q4	3	23,000	3	Q4 -> Season 3
2	Q1	4	34,000	4	Q1 -> Season 4
2	Q2	5	10,000	1	
2	Q3	6	18,000	2	
2	Q4	7	23,000	3	
3	Q1	8	38,000	4	
3	Q2	9	12,000	1	
3	Q3	10	13,000	2	
3	Q4	11	32,000	3	
4	Q1	12	41,000	4	

Note: This table shows quarterly demand data for Tahoe Salt Co. over 4 years (12 periods). The data will be used to forecast demand for periods 13-16.

7.3. Time-Series Forecasting Methods

Example: Tahoe Salt Co. (Quarterly Demand)

We will forecast the demand of Tahoe Salt Co. for period 13 - 16 using historical data from periods 1-12.



7.3. Time-Series Forecasting Methods

Estimating level and trend - Deseasonalizing demand data

Before estimating level and trend, demand data must be deseasonalized to remove seasonal fluctuations and reveal the underlying pattern.

Key Concepts

Deseasonalized demand: The demand that would have been observed in the absence of seasonal fluctuations

Periodicity (p): The number of periods after which the seasonal cycle repeats itself

For Tahoe Salt Co., p = 4 (quarterly data, seasonal cycle repeats every 4 periods)

> **Key insight:** Deseasonalizing demand helps identify the true level and trend by removing predictable seasonal variations.

7.3. Time-Series Forecasting Methods

Deseasonalizing demand

Mathematical formulas for removing seasonal effects from demand data

For p even

Periodicity is even number

$$\bar{D}_t = \left[D_{t-(p/2)} + D_{t+(p/2)} + \sum_{i=t+1-(p/2)}^{t-1+(p/2)} 2D_i \right] / 2p$$

For p odd

Periodicity is odd number

$$\bar{D}_t = \sum_{i=t-[(p-1)/2]}^{t+[(p-1)/2]} D_i / p$$

Note: Deseasonalized demand represents the demand that would have been observed in the absence of seasonal fluctuations. The formula calculates a centered moving average to remove seasonal patterns and reveal the underlying trend.

7.3. Time-Series Forecasting Methods

Tahoe Salt Co.: Deseasonalizing demand

Example calculation for t=3 with p=4 (even case) and comparison with p=3 (odd case)

Tahoe Salt Co. Quarterly Demand

Year	Quarter	Period t	Demand Dt	Season ID	Season mapping note	Decenteralized demand
1	Q2	1	8,000	1	Q2 -> Season 1	
1	Q3	2	13,000	2	Q3 -> Season 2	
1	Q4	3	23,000	3	Q4 -> Season 3	19750
2	Q1	4	34,000	4	Q1 -> Season 4	20625
2	Q2	5	10,000	1		21250
2	Q3	6	18,000	2		21750
2	Q4	7	23,000	3		22500
3	Q1	8	38,000	4		22125
3	Q2	9	12,000	1		22625
3	Q3	10	13,000	2		24125
3	Q4	11	32,000	3		
4	Q1	12	41,000	4		

=(D4+D8+2*SUM(D5:D7))/8

Tahoe Salt Co. Quarterly Demand

d t	Demand Dt	Season ID	Season mapping note	Decenteralized demand
1	8,000	1	Q2 -> Season 1	
2	13,000	2	Q3 -> Season 2	
3	23,000	3	Q4 -> Season 3	=(D4+D8+2*SUM(D5:D7))/8
4	34,000	4	Q1 -> Season 4	20625
5	10,000	1		21250
6	18,000	2		21750

For t=3, p=4 (even)

Quarterly data, even periodicity

1. D3 = [D3-2 + D3+2 + Σ2Di] / 2p
2. D3 = [D1 + D5 + 2D2 + 2D3 + 2D4] / 8
3. D3 = [8,000 + 10,000 + 2*13,000 + 2*23,000 + 2*34,000]/8
4. D3 = 158,000 / 8 = **19,750**

For t=3, p=3 (odd)

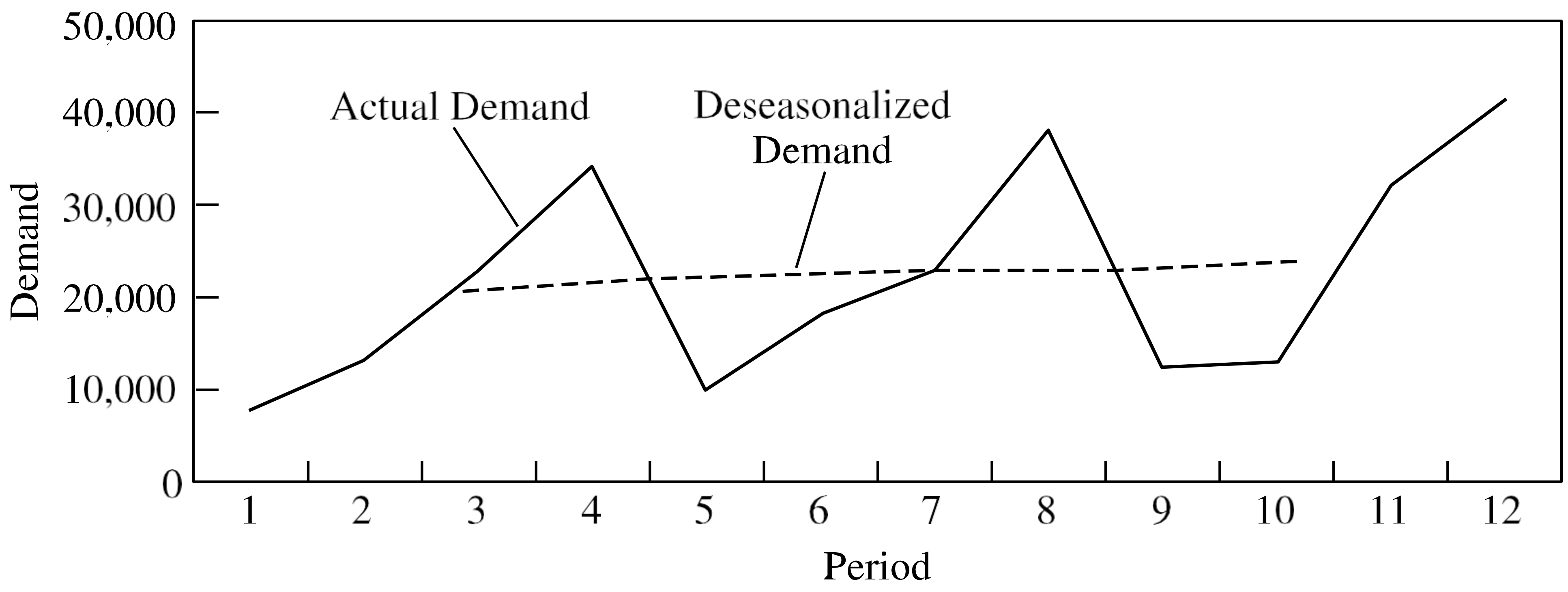
Quarterly data, odd periodicity

1. D3 = ΣDi / p
2. D3 = [D2 + D3 + D4] / 3
3. D3 = [13,000 + 23,000 + 34,000] / 3
4. D3 = 70,000 / 3 = **23,333**

7.3. Time-Series Forecasting Methods

Forecasting methods for demand prediction and trend analysis

This section covers various time-series forecasting methods including moving average, exponential smoothing, and adaptive forecasting techniques for demand prediction.



Note: Time-series forecasting uses historical data to predict future demand patterns and trends.

7.3. Time-Series Forecasting Methods

Forecasting methods for demand prediction and trend analysis

Tahoe Salt Co.: Deseasonalizing demand (estimating L and T)

The linear relationship between $\bar{D}_t$ and time t:

$$\bar{D}_t = L + tT$$

Where:
- $\bar{D}_t$ = deseasonalized demand in period t
- **L** = level (deseasonalized demand at period 0)
- **T** = trend rate of growth of deseasonalized demand)

Data| Data Analysis | Regression in Excel

$$T = \frac{\sum t\bar{D}_t - n\bar{\bar{D}}_{\bar{t}}}{\sum t^2 - n\bar{t}^2} \qquad L = \bar{\bar{D}} - T\bar{t}$$

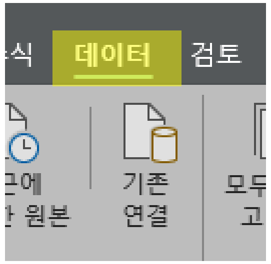
Note: Trend is determined **by linear regression** using deseasonalized demand as the dependent variable and period as the independent variable

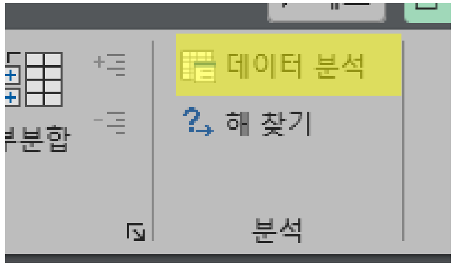
In the example, L = 18,439 and T = 524, so $\bar{D}_t = 18{,}439 + 524t$

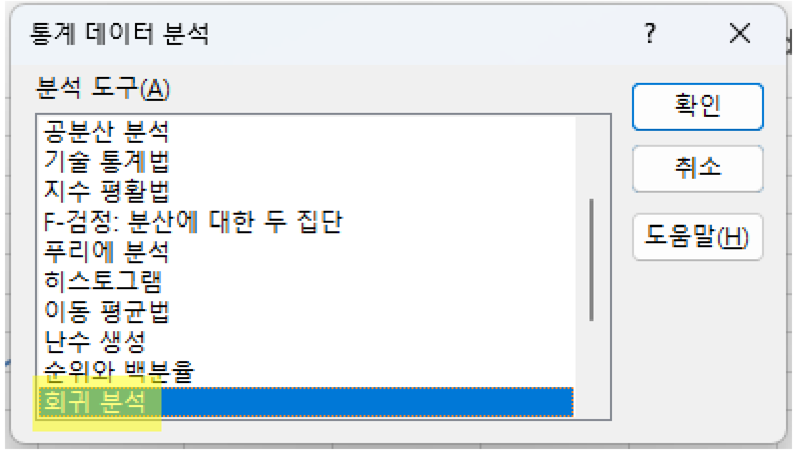
7.3. Time-Series Forecasting Methods

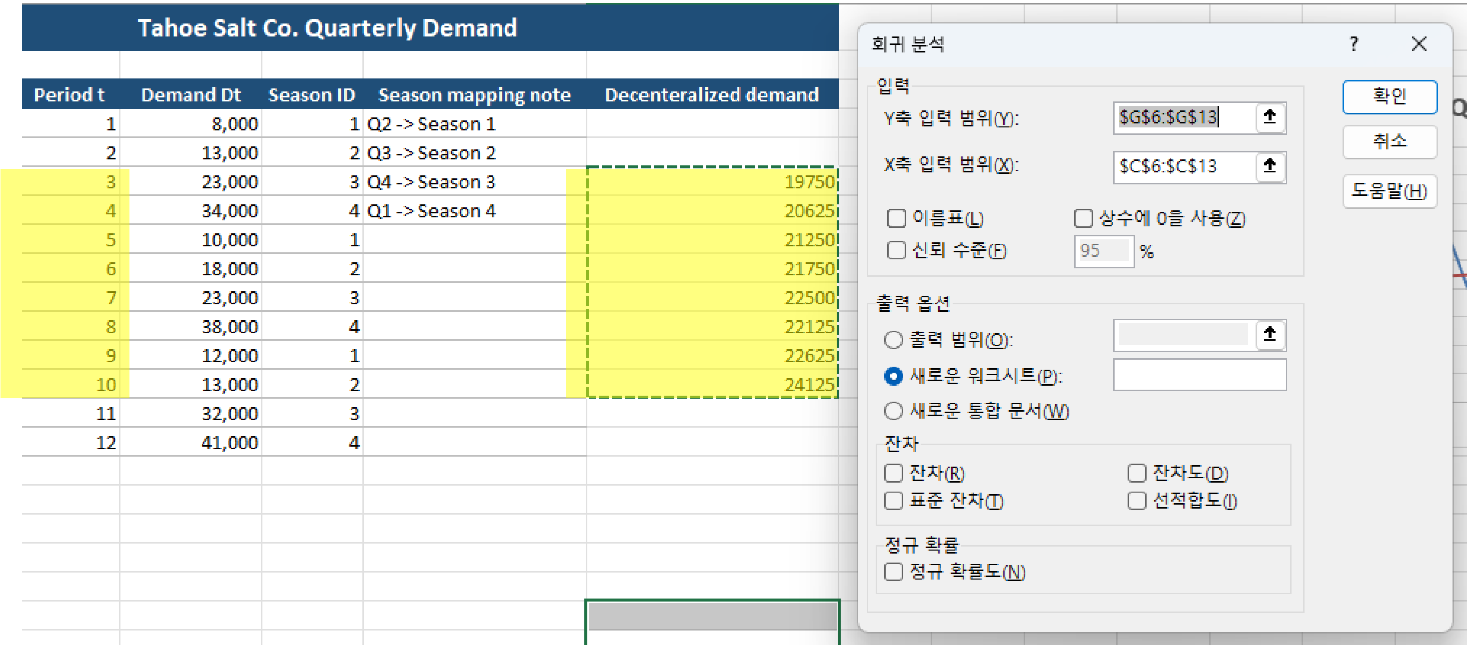
Forecasting methods for demand prediction and trend analysis

Tahoe Salt Co.: Deseasonalizing demand (estimating L and T)









Tahoe Salt Co. Quarterly Demand

Period t	Demand Dt	Season ID	Season mapping note	Decenteralized demand
1	8,000	1	Q2 -> Season 1	
2	13,000	2	Q3 -> Season 2	
3	23,000	3	Q4 -> Season 3	19750
4	34,000	4	Q1 -> Season 4	20625
5	10,000	1		21250
6	18,000	2		21750
7	23,000	3		22500
8	38,000	4		22125
9	12,000	1		22625
10	13,000	2		24125
11	32,000	3		
12	41,000	4		

7.3. Time-Series Forecasting Methods

Forecasting methods for demand prediction and trend analysis

Tahoe Salt Co.: Deseasonalizing demand (estimating L and T)

	A	B	C	D	E	F	G	H	I	J
1	요약 출력									
2										
3	회귀분석 통계량									
4	다중 상관	0.958065								
5	결정계수	0.917889								
6	조정된 결	0.904204								
7	표준 오차	414.5033								
8	관측수	8								
9										
10	분산 분석									
11		자유도	제곱합	제곱 평균	F 비	유의한 F				
12	회귀	1	11523810	11523810	67.07182	0.000179				
13	잔차	6	1030878	171813						
14	계	7	12554688							
15										
16		계수	표준 오차	t 통계량	P-값	하위 95%	상위 95%	하위 95.0%	상위 95.0%	
17	Y 절편	18438.99	440.8087	41.82991	1.25E-08	17360.37	19517.61	17360.37	19517.61	
18	X 1	523.8095	63.95925	8.189738	0.000179	367.3069	680.3122	367.3069	680.3122	
19										

$$T = \frac{\sum t\bar{D}_t - n\bar{\bar{D}}_{\bar{t}}}{\sum t^2 - n\bar{t}^2}$$

$$L = \bar{\bar{D}} - T\bar{t}$$

In the example, L = 18,439 and T = 524, so $\bar{D}_t = 18{,}439 + 524t$

7.3. Time-Series Forecasting Methods

Forecasting methods for demand prediction and trend analysis

Tahoe Salt Co.: Deseasonalizing demand (estimating L and T)

Use deseasonalized demand, $\overline{D}'_t$, obtained by using the previous equation to **compute the seasonal factors for period t**

$$\overline{D}_t = 18{,}439 + 524t$$

$$\bar{S}_t = D_t/\overline{D'}_t$$

where Dt is the actual demand

For instance,

$$\overline{D}'_2 = 18{,}439 + (524)(2) = 19{,}487,\ D_2 = 13{,}000$$
$$\overline{S}_2 = 13{,}000/19{,}487 = 0.67$$

The seasonal factors for the other periods are calculated in the same manner

7.3. Time-Series Forecasting Methods

Tahoe Salt Co.: Deseasonalized demand and seasonal factors

Combined table showing deseasonalized demand values and seasonal factors for trend analysis

Deseasonalized Demand

Demand values after removing seasonal effects

Period t	Demand Dt	Deseasonalized demand D't	St
1	8,000	18963	0.42
2	13,000	19487	0.67
3	23,000	20011	1.15
4	34,000	20535	1.66
5	10,000	21059	0.47
6	18,000	21583	0.83
7	23,000	22107	1.04
8	38,000	22631	1.68
9	12,000	23155	0.52
10	13,000	23679	0.55
11	32,000	24203	1.32
12	41,000	24727	1.66

Seasonal Factors

Calculated seasonal indices for each quarter

$$\overline{D}_t = 18{,}439 + 524t$$

$$\overline{S_t} = D_t/\overline{D}'_t$$

Period t	Demand Dt	Deseasonalized demand D't
1	8,000	=18439+524*A4
2	13,000	19487

	Period t	Demand Dt	Deseasonalized demand D't	St
3				
4	1	8,000	18963	=B4/C4
5	2	13,000	19487	0.67

7.3. Time-Series Forecasting Methods

Tahoe Salt Co.: Estimating seasonal factors

Calculating seasonal factors by averaging deseasonalized demand across cycles

Seasonal Factor Calculation

Averaging method for seasonal indices

$$S_i = \sum_{j=0}^{r-1} \bar{S}_{j \cdot p+i} \, / r$$

Where r = number of seasonal cycles

Si = Average of all seasonal factors for season i

The overall seasonal factor for a "season" is obtained by averaging all factors for that season across all cycles.

Period t	Demand Dt	Deseasonalized demand D't	St	Quarter	Season ID	Ft
1	8,000	18963	0.4219	Q2	1	
2	13,000	19487	0.6671	Q3	2	
3	23,000	20011	1.1494	Q4	3	
4	34,000	20535	1.6557	Q1	4	
5	10,000	21059	0.4749	Q2	1	
6	18,000	21583	0.8340	Q3	2	
7	23,000	22107	1.0404	Q4	3	
8	38,000	22631	1.6791	Q1	4	
9	12,000	23155	0.5182	Q2	1	
10	13,000	23679	0.5490	Q3	2	
11	32,000	24203	1.3222	Q4	3	
12	41,000	24727	1.6581	Q1	4	

s1	=(D4+D8+D12)/3
s2	0.68
s3	1.17
s4	1.66
sum	3.99

Calculated Seasonal Factors For Tahoe Salt Co. (p=4, r=3 cycles)

Quarter	Season	Factor	Value
Q2	S1	(0.42+0.47+0.52)/3	**0.47**
Q3	S2	(0.67+0.83+0.55)/3	**0.68**
Q4	S3	(1.15+1.04+1.32)/3	**1.17**
Q1	S4	(1.66+1.68+1.66)/3	**1.67**

Verification

Sum of seasonal factors

Sum of factors: 0.47 + 0.68 + 1.17 + 1.67 = **3.99** ≈ 4.0

(Should equal number of periods in cycle)

7.3. Time-Series Forecasting Methods

Tahoe Salt Co.: Estimating the forecast

Using the mixed model equation to forecast demand for periods 13-16 with L=18,439 and T=524

Mixed Model Formula

$$F_t = (L + tT) \times S_t$$

Where:
- **L** = 18,439 (estimate of level for Period 0)
- **T** = 524 (estimate of trend)
- **S_t** = seasonal factor for period t

Seasonal Factors

- S_1 = 0.47 (Q2)
- S_2 = 0.68 (Q3)
- S_3 = 1.17 (Q4)
- S_4 = 1.67 (Q1)

Detailed Calculation Steps

F_{13}: (18,439 + 6,812) × 0.47 = 25,251 × 0.47 = **11,868**

F_{14}: (18,439 + 7,336) × 0.68 = 25,775 × 0.68 = **17,527**

F_{15}: (18,439 + 7,860) × 1.17 = 26,299 × 1.17 = **30,770**

F_{16}: (18,439 + 8,384) × 1.67 = 26,823 × 1.67 = **44,794**

1 F13 (Q2)
11,868
(18,439 + 13×524) × 0.47

2 F14 (Q3)
17,527
(18,439 + 14×524) × 0.68

3 F15 (Q4)
30,770
(18,439 + 15×524) × 1.17

4 F16 (Q1)
44,794
(18,439 + 16×524) × 1.67

7.3. Time-Series Forecasting Methods

Tahoe Salt Co.: Estimating the forecast

Using the mixed model equation to forecast demand for periods 13-16 with L=18,439 and T=524

1 F13 (Q2)
11,868
(18,439 + 13×524) × 0.47

2 F14 (Q3)
17,527
(18,439 + 14×524) × 0.68

3 F15 (Q4)
30,770
(18,439 + 15×524) × 1.17

4 F16 (Q1)
44,794
(18,439 + 16×524) × 1.67

Period t	Demand Dt	Deseasonalized demand D't	St	Quarter	Season ID	Ft
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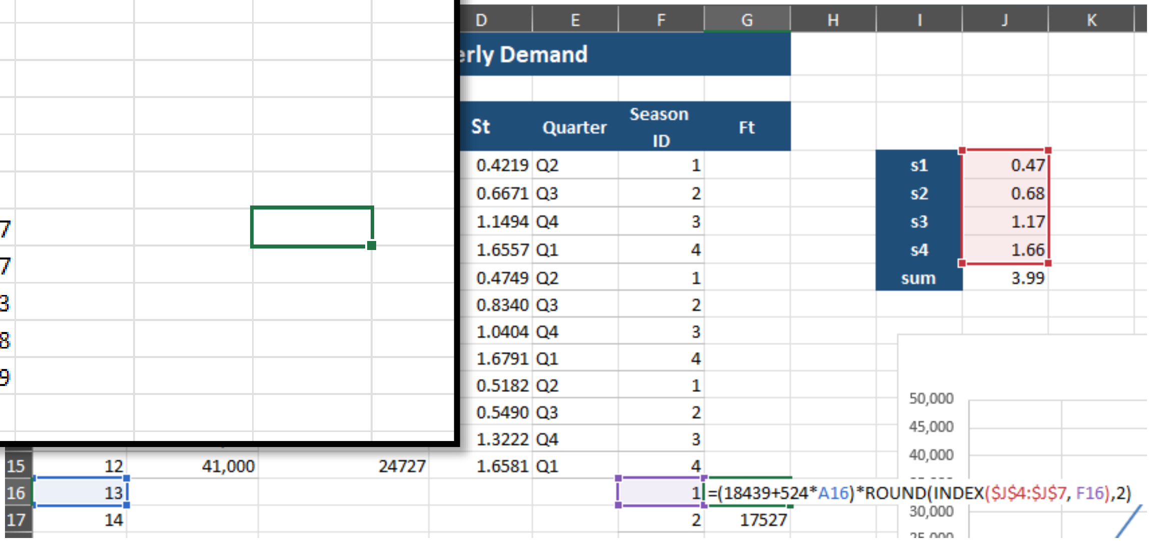
7	23,000	22107	1.0404	Q4	3	
8	38,000	22631	1.6791	Q1	4	
9	12,000	23155	0.5182	Q2	1	
10	13,000	23679	0.5490	Q3	2	
11	32,000	24203	1.3222	Q4	3	
12	41,000	24727	1.6581	Q1	4	
13					1	11867.97
14					2	17527
15					3	30769.83
16					4	44526.18
17					1	12853.09

s1	0.47
s2	0.68
s3	1.17
s4	1.66
sum	3.99



7.3. Time-Series Forecasting Methods

Adaptive forecasting

Estimates of level, trend, and seasonality are updated after each demand observation

Adaptive forecasting involves continuously updating the estimates of level, trend, and seasonality after each new demand observation. This allows the forecast to adapt to changing conditions and improve accuracy over time.

The key advantage is that the model learns from new data and adjusts its parameters accordingly.

Moving average Level only

Used when demand has no observable trend or seasonality. The level is updated as the average of the last N periods.

Simple exponential smoothing Level only

Uses exponential weights to give more importance to recent observations. Best for stable demand patterns.

Trend-corrected exponential smoothing (Holt's model) Level + Trend

Appropriate when demand has level and trend components, but no seasonality.

Trend- and seasonality-corrected exponential smoothing (Winter's model) Level + Trend + Seasonality

Most comprehensive model for demand with all three components: level, trend, and seasonality.

Key insight: Adaptive methods continuously update forecasts based on new data, making them more responsive to changes in demand patterns.

7.3. Time-Series Forecasting Methods

Moving average method - Used when demand has no observable trend or

When to Use

Used when demand has no observable trend or seasonality. The systematic component of demand is only level.

Systematic Component

The level in period t is the average demand over the last N periods (i.e., the N-period moving average).

Moving Average Formula

$$L_t = (D_t + D_{t-1} + \ldots + D_{t-N+1}) / N$$

Where Lt = Level in period t, Di = Demand in period i, N = Number of periods

- **L_t (Level)**
  Average demand over the last N periods
- **D_i (Demand)**
  Actual demand observed in period i
- **N (Periods)**
  Number of periods for averaging

Example: If N=4 and demand for the last 4 periods is [80, 90, 85, 95], then L_t = (80+90+85+95)/4 = 87.5

Example 7.1: Moving average

Weekly Sales Forecast of "wheat cereals in Great Britain"

Problem Statement

Sales over the last four weeks of April: **38, 35, 77, 90 thousand tons**. What is the forecast demand for the 1st and 2nd weeks of May using a four-period (N = 4) moving average?

Analysis: Calculations

$L_4 = (38 + 35 + 77 + 90) / 4 = 60$

$F_5 = L_4 = 60$ thousand tons

$D_5 = 80$ (observed)

$E_5 = F_5 - D_5 = 60 - 80 = -20$

$L_5 = (35 + 77 + 90 + 80) / 4 = 70.5$

$F_6 = L_5 = 70.5$ thousand tons

Forecast Results

- Forecast for 1st week of May: **60 thousand tons**
- Forecast for 2nd week of May: **70.5 thousand tons**

Key Insight: The moving average smooths out fluctuations and provides a stable forecast.

7.3. Time-Series Forecasting Methods

Simple Exponential Smoothing

Usage: Simple exponential smoothing is used when demand has **no observable trend or seasonality** (So, the systematic component of demand is only level). The level in period t is updated based on the observed demand and the previous level estimate.

Initial Estimate

$$L_0 = (D_1 + D_2 + ... + D_n) / n = F_1$$

L_0 is assumed to be the **average of all historical data**

Update Formula

$$L_t = \alpha D_t + (1-\alpha) L_{t-1} = F_t + \alpha (D_t - F_t)$$

After observing demand $\mathbf{D_t}$, update level estimate $\mathbf{L_t}$

Exponentially Increased Weights to Recent Observations

The simple exponential smoothing method gives **exponentially increasing weights to recent observations**. The smoothing constant α ($0 < \alpha \leq 1$) determines how much weight is given to the most recent demand observation versus the previous forecast.

Exponential Smoothing Formula:

$$F_{t+1} = L_t = \alpha D_t + (1-\alpha) L_{t-1}$$

Key properties:

- When α is close to 1, more weight is given to recent demand
- When α is close to 0, more weight is given to the previous forecast

7.3. Time-Series Forecasting Methods

Simple exponential smoothing method - Exponentially increased weights to recent observations

Exponential Smoothing Formula

The forecast for period t+1 is calculated as:

$$F_{t+1} = \alpha D_t + (1-\alpha) F_t$$

Where:

- **F_{t+1}** = Forecast for next period
- **D_t** = Actual demand in period t
- **α** = Smoothing constant (0 < α < 1)

Mathematical Proof

Why weights sum to 1:

The sum of all weights is a geometric series: α + α(1-α) + α(1-α)² + ... = α/(1-(1-α)) = 1

Exponential Smoothing Concept

$$F_{t+1} = \alpha D_t + (1-\alpha) F_t$$

$$F_t = \alpha D_{t-1} + (1-\alpha) F_{t-1}$$

$$F_{t-1} = \alpha D_{t-2} + (1-\alpha) F_{t-2}$$

$$F_{t+1} = \alpha D_t + \alpha(1-\alpha)D_{t-1} + \alpha(1-\alpha)^2 D_{t-2} + \cdots + \alpha(1-\alpha)^{t-1}D_1 + (1-\alpha)^t F_1$$

7.3. Time-Series Forecasting Methods

Example 7.2: Simple Exponential Smoothing

Weekly Sales Forecast of "wheat cereals in Great Britain" using α=0.1

Problem Statement

What is the sales forecast for the **1st and 2nd weeks of May** using simple exponential smoothing method with **α = 0.1**? Sales over the last four weeks of April: **38, 35, 77, 90 thousand tons**.

Analysis: Calculations

$(38 + 35 + 77 + 90) / 4 = 60$

$L_0 = 60$ (initial forecast)

38 (observed)

$0.1 \times 38 + 0.9 \times 60 = 57.8$

$L_1 = 57.8$

Forecast Results

- Forecast for 1st week of May: **57.8 thousand tons**
- Forecast for 2nd week of May: **56.02 thousand tons**

Key Insight: The forecast gradually adjusts based on the smoothing constant α=0.1.

Step 1
Calculate initial level L_0

Step 2
Apply smoothing formula

Step 3
Update level estimate

Step 4
Generate forecast

7.3. Time-Series Forecasting Methods

Trend-Corrected Exponential Smoothing (Holt's Model)

Appropriate when demand has a level and trend but no seasonality

Holt's Model Overview

Holt's model is used when demand has both **level** and **trend** components in the systematic part of demand, but **no seasonality**. The model separately estimates level and trend using exponential smoothing.

> **Key Feature:** The model can adapt to changes in both the level and trend of demand over time.

Initial Estimates

Obtain initial estimates of level (L_0) and trend (T_0) by running a **linear regression** of the form $D_t = at + b$. Then:

- $\mathbf{T_0 = a}$ (slope from regression)
- $\mathbf{L_0 = b}$ (intercept from regression)
- For Period 1: $\mathbf{F_1 = L_0 + T_0}$

Update Formulas

After observing demand D_t in period t, update estimates:

$$L_t = \alpha D_t + (1-\alpha)(L_{t-1} + T_{t-1})$$

$$T_t = \beta(L_t - L_{t-1}) + (1-\beta)T_{t-1}$$

$$F_{t+1} = L_t + T_t$$

α (Level smoothing)

$0 < \alpha \leq 1$

β (Trend smoothing)

$0 < \beta \leq 1$

Forecast Calculation

For Period t+1: $\mathbf{F_{t+1} = L_t + T_t}$ (level + trend)

7.3. Time-Series Forecasting Methods

Example 7.3: Holt's model

Japan National Tourist Organization: Visitors forecast to Japan using linear regression for 2002-2007

Problem Statement

Using linear regression for the number of visitors during the year of 2002-2007, with L_0 **= 2,604,842** and T_0 **= 548,247**. Forecast for year 2008?

Year	Visitors
2002	2,604,842
2003	3,153,089
2004	3,701,336
2005	4,233,360
2006	4,780,000
2007	5,328,247

Level (L_0)
2,604,842

Trend (T_0)
548,247

Smoothing α
0.2 (for level)

Smoothing β
0.1 (for trend)

Forecast Calculation

Using Holt's model with trend correction:

$$F_{t+1} = L_t + T_t$$

$$L_t = \alpha D_t + (1-\alpha)(L_{t-1} + T_{t-1})$$
$$T_t = \beta(L_t - L_{t-1}) + (1-\beta)T_{t-1}$$

Forecast for Period 1 (or year 2002):

$F_1 = L_0 + T_0 = 3{,}153{,}809$

Observed number for Period 1 (D_1 = 3,417,774), so

$E_1 = F_1 - D_1 = -264{,}685$

With $\alpha = 0.1$, $\beta = 0.2$,

$L_1 = \alpha D_1 + (1 - \alpha)(L_0 + T_0) = 3{,}179{,}558$

$T_1 = \beta(L_1 - L_0) + (1 - \beta)T_0 = 553{,}541$

So, $F_2 = L_1 + T_1 = 3{,}179{,}558 + 553{,}541 = 3{,}733{,}099$ visitors

Continuing in this manner, F_7 (year 2008) $= L_6 + T_6 = 6{,}439{,}353$

For 2008 (t=7):

7.3. Time-Series Forecasting Methods

Example 7.3: Holt's model

Japan National Tourist Organization: Visitors forecast to Japan using linear regression for 2002-2007

Trend-Corrected Exponential Smoothing (Holt's Model)

Input / As	Value		Formulas
Alpha (α)	0.1		F1 = L0 + T0
Beta (β)	0.2		Et = Ft - Dt
Initial Level I	2,604,842		Lt = αDt + (1-α)(L(t-1)+T(t-1))
Initial Trend	548,247		Tt = β(Lt-L(t-1)) + (1-β)T(t-1)

Period t	Year	Actual Demand Dt	Forecast Ft	Error Et	Level Lt	Trend Tt	Forecast for next period	Remarks
1	2002	3,417,774	3,153,089	-264,685	3,179,558	553,541	3,733,099	Uses initial L0 and T0
2	2003	3,511,513	3,733,099	221,586	3,710,940	549,109	4,260,050	
3	2004	4,208,095	4,260,050	51,955	4,254,854	548,070	4,802,924	
4	2005	4,627,478	4,802,924	175,446	4,785,380	544,561	5,329,941	
5	2006	5,247,125	5,329,941	82,816	5,321,660	542,905	5,864,565	
6	2007	6,130,262	5,864,565	-265,697	5,891,134	548,219	6,439,353	
7	2008		6,439,353				6,439,353	Forecast for 2008

7.3. Time-Series Forecasting Methods

Forecasting confidence interval

Statistical analysis of forecast errors and confidence interval calculations

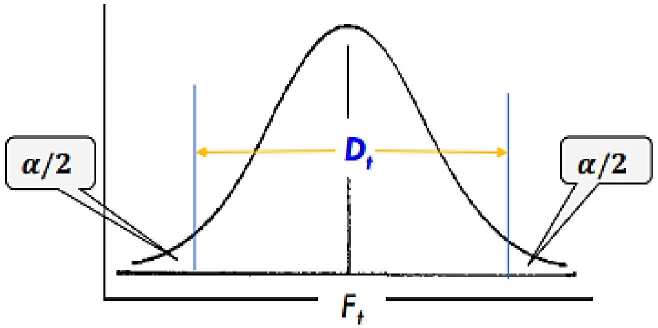
Forecast Error Distribution

Under normal circumstances, forecast errors are **normally distributed** as **NORM(0, σ)**, where σ represents the standard deviation of the forecast errors. This assumption allows us to construct confidence intervals for our forecasts.

95% Confidence Interval: $F_{t+1} \pm 1.96\sigma$

Key Insight: The confidence interval widens as the forecast horizon increases, reflecting greater uncertainty in long-term predictions.

$$
\begin{aligned}
D_t &= F_t \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\\
&= 1{,}064 \pm z_{0.025}(408)\\
&= 1{,}064 \pm 1.96(408)\\
&= 1{,}064 \pm 800\\
\therefore\ & 264 \le D_t \le 1{,}864
\end{aligned}
$$



95% Confidence Interval Calculation

$$F_t - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \le D_t \le F_t + z_{\alpha/2}\frac{\sigma}{\sqrt{n}};$$

$$\text{where } \sigma = \sqrt{\frac{\sum_{t=1}^{n}(F_t - D_t)^2}{n-1}}$$

For a 95% confidence level (α = 0.05):

Formula Application

If the forecast Ft+1 = 100,000 and σ = 5,000, then:

95% CI: 100,000 ± 1.96 × 5,000 = [90,200, 109,800]

Interpretation

We are 95% confident that the actual demand will fall within the calculated interval.

7.3. Time-Series Forecasting Methods

Combined model forecasting

Clothing Manufacture's Seasonal Forecasting: Four models weighted by forecast error

Combined Forecasting Concept

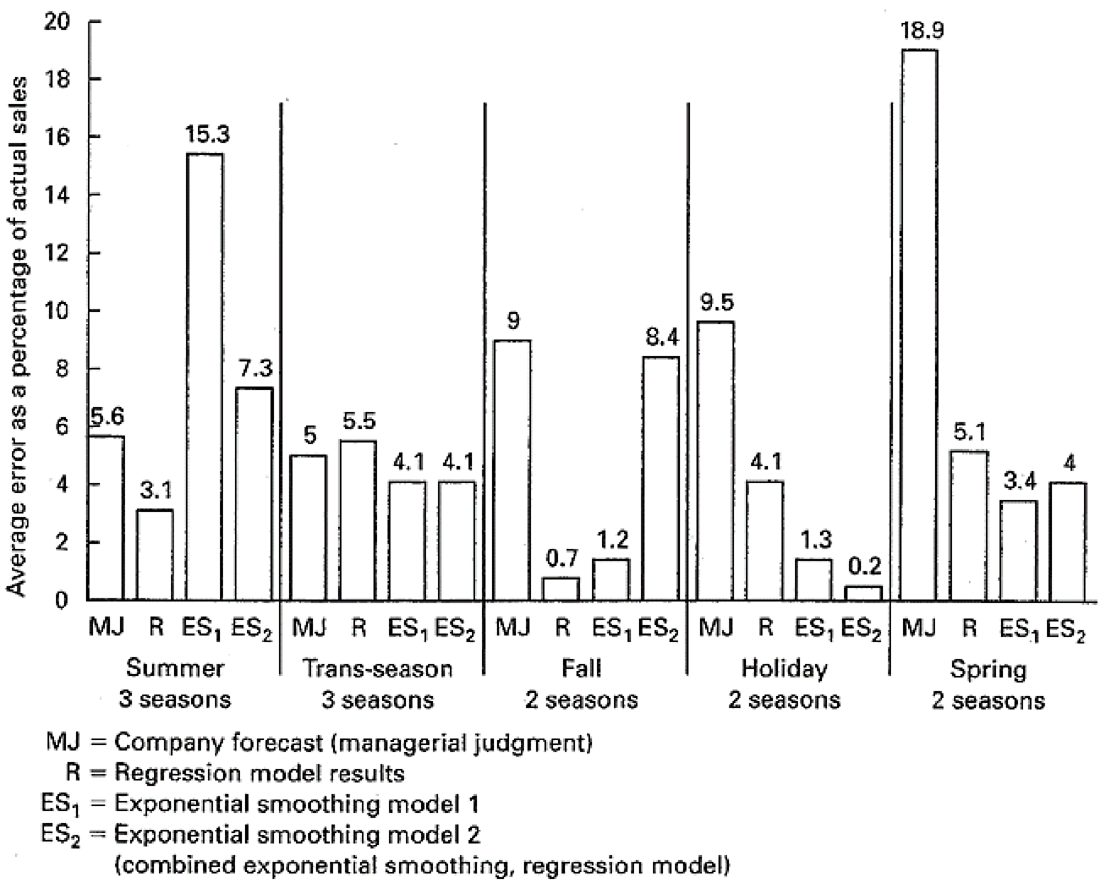
When multiple forecasting models are available, we can combine them by weighting each forecast according to its respective forecast error. Models with lower errors receive higher weights.

Key Principle: The combined forecast is a weighted average of individual model forecasts, where weights are inversely proportional to forecast errors.

Weight Calculation

Weight for model i: $w_i = (1/MSE_i) / \Sigma(1/MSE_j)$

Combined Forecast = $\Sigma(w_i \times Forecast_i)$



7.3. Time-Series Forecasting Methods

Combined model forecasting

This section demonstrates how to combine multiple forecasting models to improve prediction accuracy by weighting forecasts according to their respective errors.

Combined Forecasting Results

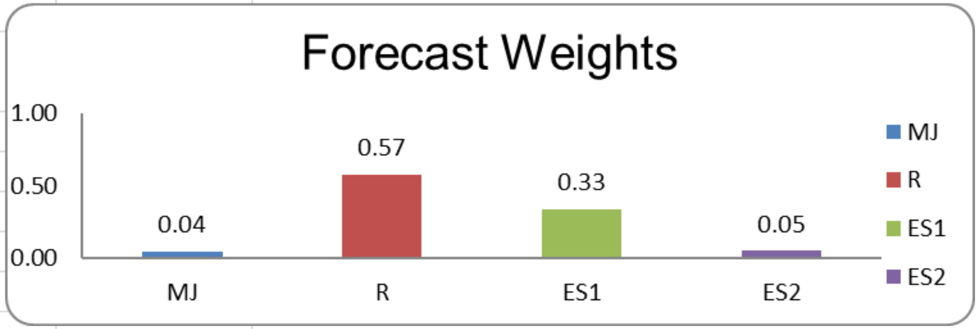
Multiple model approach for improved forecasting accuracy

Forecast weight calculation

그림 영역 el	Forecast error	Percent total error	Inverse error proportion	Model weight	Rounded weight
MJ	9.00	0.47	2.14	0.04	0.04
R	0.70	0.04	27.57	0.57	0.58
ES1	1.20	0.06	16.08	0.33	0.33
ES2	8.40	0.44	2.30	0.05	0.05
Total	19.30		48.10		1.00

Weighted average forecast

Forecast type	Model forecast	Weight	Weighted proportion
Regression model (R)	20,367,000	0.58	11,813,000
Exponential smoothing (ES1)	20,400,000	0.33	6,732,000
Combined exp-smoothing-regression (ES2)	17,660,000	0.05	883,000
Managerial judgment (MJ)	19,500,000	0.04	780,000
Weighted average forecast			20,208,000



7.4. Measures of Forecast Error

Statistical measures to evaluate forecast accuracy and model performance

Forecast Error

The difference between forecast and actual demand:

$$E_t = F_t - D_t$$

Where Et = forecast error, Ft = forecast, Dt = actual demand

Mean Squared Error (MSE)

Average of squared forecast errors:

$$MSE = (1/n) \times \Sigma(E_t)^2$$

Penalizes large errors more heavily

Mean Absolute Deviation (MAD)

Average of absolute forecast errors:

$$MAD = (1/n) \times \Sigma|E_t|$$

More robust to outliers than MSE

Mean Absolute % Error (MAPE)

Percentage error relative to actual:

$$MAPE = (100/n) \times \Sigma|E_t/D_t|$$

Useful for comparing across different scales

Bias

Systematic forecast error (should be ~0):

$$Bias = (1/n) \times \Sigma E_t$$

Note: Should fluctuate around 0

Tracking Signal (TS)

Ratio of cumulative error to MAD:

$$TS = \Sigma E_t / MAD$$

Monitors forecast model adequacy

7.4. Measures of Forecast Error

Tracking signal control chart

Monitoring forecast model adequacy with tracking signal analysis

Tracking Signal Concept

The **tracking signal (TS)** monitors the fit of the forecasting model by tracking the ratio of cumulative forecast error to mean absolute deviation (MAD).

Key Formula: TS t = BIAS t / MAD t

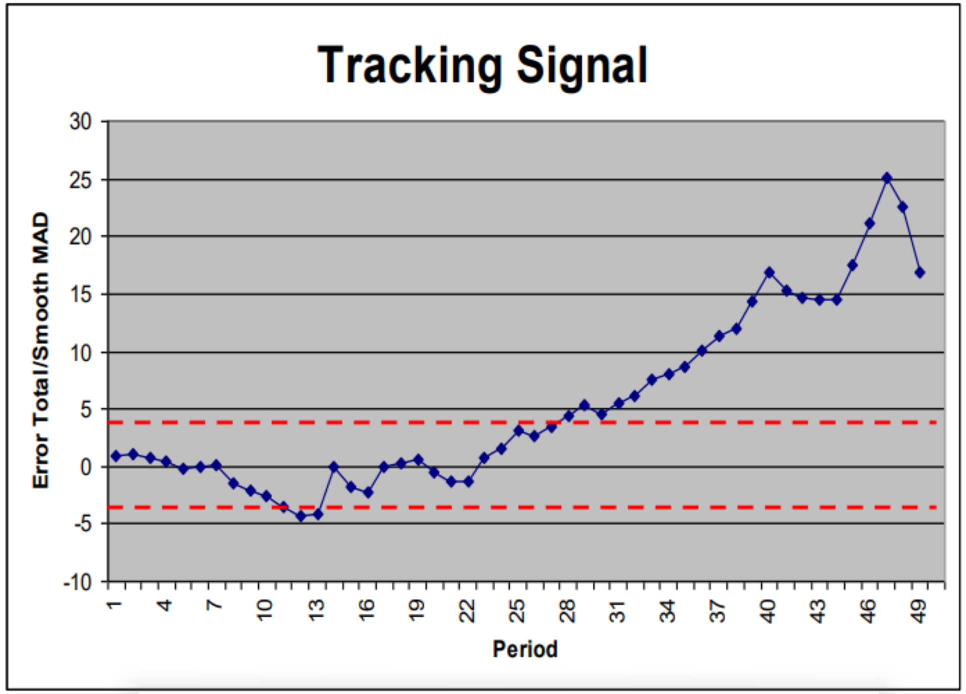
- Center line = 0 (ideal forecast)
- Upper control limit = +3 to +5
- Lower control limit = -3 to -5

Signal Interpretation

If TS falls outside control limits, it indicates the forecasting model is no longer adequate and needs revision.

- Within limits: Model is adequate

Tracking Signal Control Chart



Upper Limit	Center Line	Lower Limit
+3 to +5	0	-3 to -5

7.5. Building Forecasting Models using Excel for Tahoe Salt Co.

Excel implementation guide for demand forecasting methods

Excel Forecasting Methods

Three essential forecasting models for Tahoe Salt Co. demand analysis

Moving Average

Simple method using historical demand averages over a specified number of periods.

- Used when demand has no observable trend or seasonality
- Systematic component is only level
- Level = average demand over last N periods

Simple Exponential Smoothing

Exponentially weighted average giving more weight to recent observations.

- Used when demand has no trend/seasonality
- Initial estimate: average of all historical data
- Updated after each demand observation

Trend-Corrected (Holt's)

Holt's model for demand with level and trend but no seasonality.

- Appropriate for demand with trend
- Estimates level and trend components
- Adaptive forecasting with smoothing constants

7.5. Building Forecasting Models using Excel for Tahoe Salt Co.

Selecting smoothing constant by minimizing MSE

Selecting smoothing constant by minimizing MSE

This slide demonstrates how to build forecasting models in Excel for Tahoe Salt Co. by selecting the optimal smoothing constant that minimizes the Mean Squared Error (MSE).

Selecting smoothing constant by minimizing MSE

Period t	Demand D_t	Level L_t	Forecast F_t	Error E_t	Squared Error	Absolute Error A_t	% Error
0		2017.9					
1	2024.0	2021.2	2017.9	-6.1	37.2	6.1	0.3%
2	2076.0	2050.8	2021.2	-54.8	3003.1	54.8	2.6%
3	1992.0	2019.0	2050.8	58.8	3462.6	58.8	3.0%
4	2075.0	2049.3	2019.0	-56.0	3134.6	56.0	2.7%
5	2070.0	2060.5	2049.3	-20.7	428.5	20.7	1.0%
6	2046.0	2052.7	2060.5	14.5	210.2	14.5	0.7%
7	2027.0	2038.8	2052.7	25.7	658.2	25.7	1.3%
8	1972.0	2002.7	2038.8	66.8	4459.2	66.8	3.4%
9	1912.0	1953.6	2002.7	90.7	8218.2	90.7	4.7%
10	1985.0	1970.6	1953.6	-31.4	985.0	31.4	1.6%
avg	2017.9	2021.5	2026.6	8.7	2459.7	42.5	2.1%
α =	0.540946389						

Solver setup

Set Target Cell:	F14
Equal To:	Min
By Changing Cells	B15
Constraint:	B15<=1
Also add:	B15>=0

Note: The goal is to find the smoothing constant (α) that minimizes the Mean Squared Error between forecasted and actual values.

7.5. Building Forecasting Models using Excel for Tahoe Salt Co.

Selecting smoothing constant by minimizing MAD (Mean Absolute Deviation)

Selecting smoothing constant by minimizing MAD (Mean Absolute Deviation)

This slide demonstrates how to select the optimal smoothing constant (α) for exponential smoothing by minimizing the Mean Absolute Deviation (MAD) of forecast errors.

	A	B	C	D	E	F	G	H
1	Selecting smoothing constant by minimizing MSE							
2	Period t	Demand D_t	Level L_t	Forecast F_t	Error E_t	Squared Error	Absolute Error A_t	% Error
3	0		2017.9					
4	1	2024.0	2019.8	2017.9	-6.1	37.2	6.1	0.3%
5	2	2076.0	2037.8	2019.8	-56.2	3152.9	56.2	2.7%
6	3	1992.0	2023.2	2037.8	45.8	2097.2	45.8	2.3%
7	4	2075.0	2039.7	2023.2	-51.8	2687.5	51.8	2.5%
8	5	2070.0	2049.4	2039.7	-30.3	916.4	30.3	1.5%
9	6	2046.0	2048.3	2049.4	3.4	11.6	3.4	0.2%
10	7	2027.0	2041.5	2048.3	21.3	454.3	21.3	1.1%
11	8	1972.0	2019.3	2041.5	69.5	4830.6	69.5	3.5%
12	9	1912.0	1985.0	2019.3	107.3	11511.1	107.3	5.6%
13	10	1985.0	1985.0	1985.0	0.0	0.0	0.0	0.0%
14	avg	2017.9	2024.3	2028.2	10.3	2569.9	39.2	2.0%
15	α =	0.319599051						
16								
17	**Solver setup**							
18	Set Target Cell:	G14						
19	Equal To:	Min						
20	By Changing Cells	B15						
21	Constraint:	B15<=1						
22								
23	Also add:	B15>=0						

Thank You!

경청해 주셔서 감사합니다!

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